



CHALMERS
UNIVERSITY OF TECHNOLOGY



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Computer Vision

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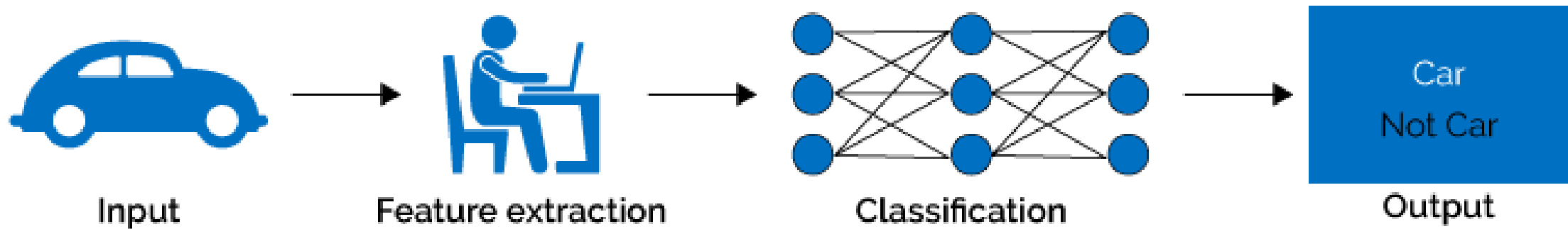
Data Science and AI

February 27, 2026

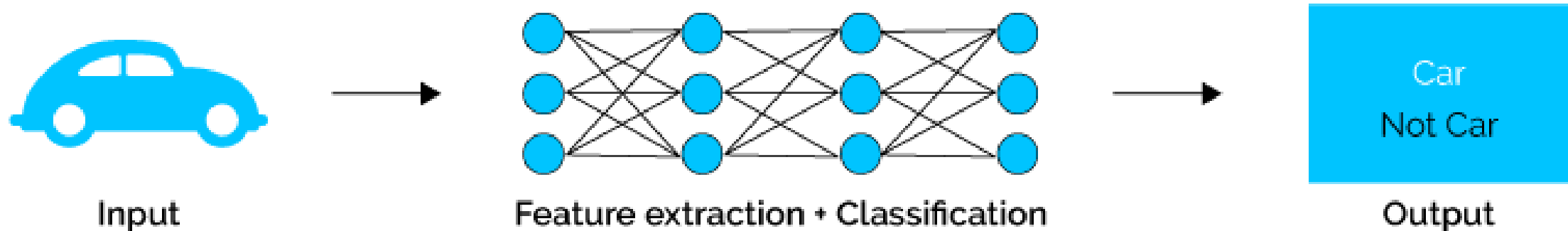
Original slides by Mohammad Kakooei

Machine learning Vs. Deep learning

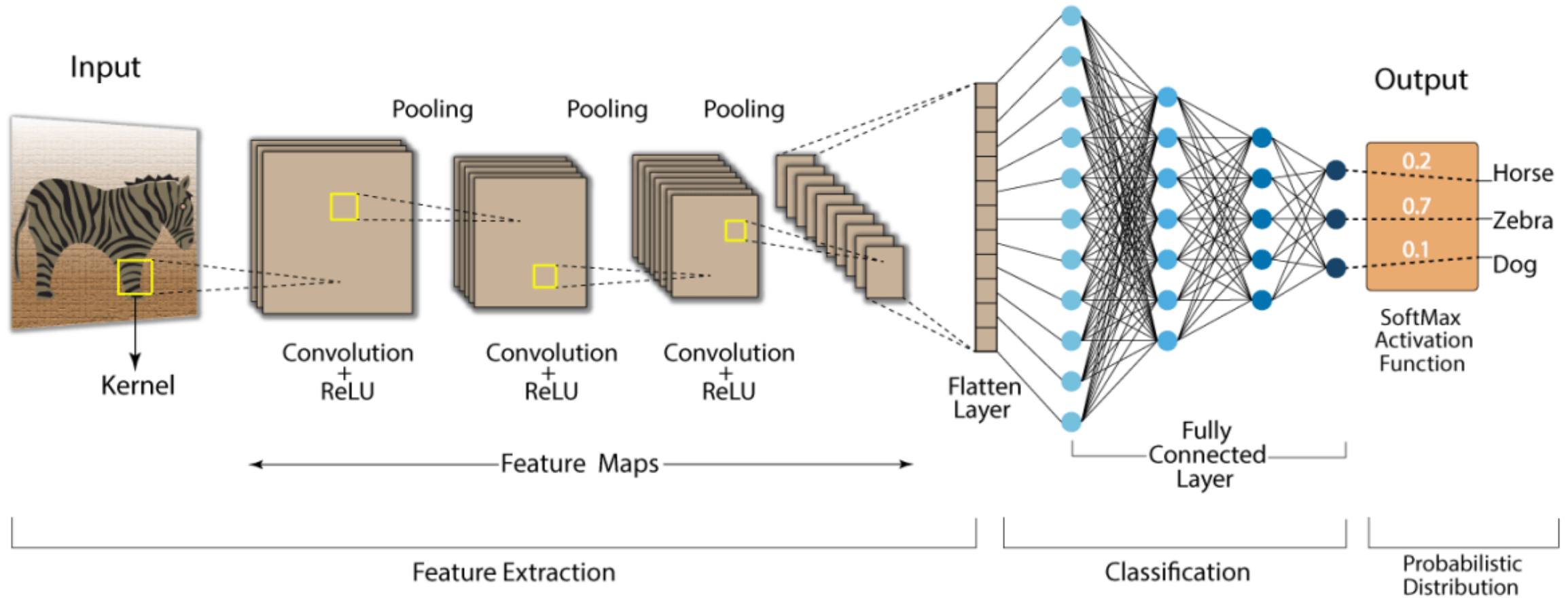
Machine Learning



Deep Learning



Sample deep network

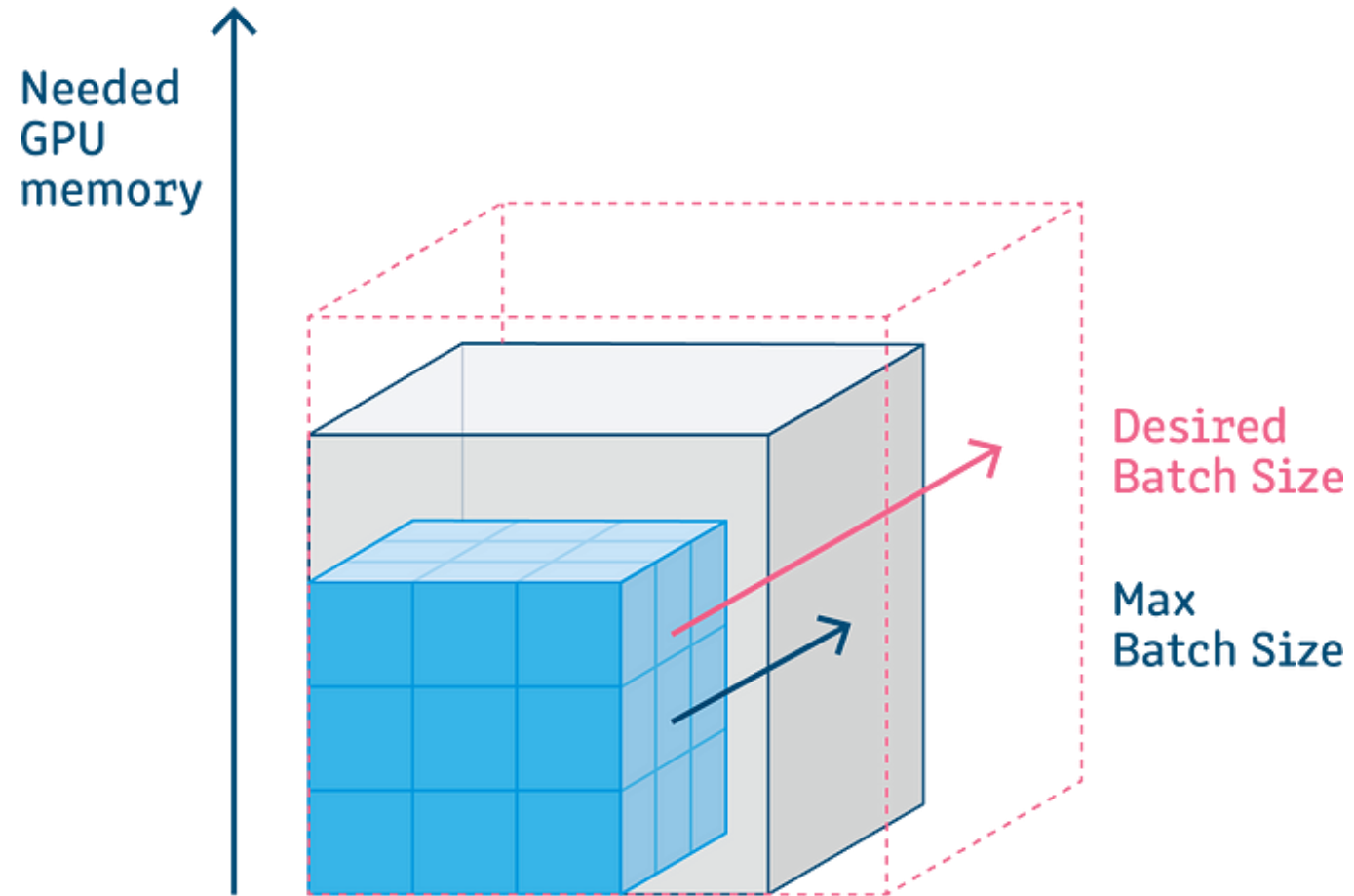


Deep learning terminology

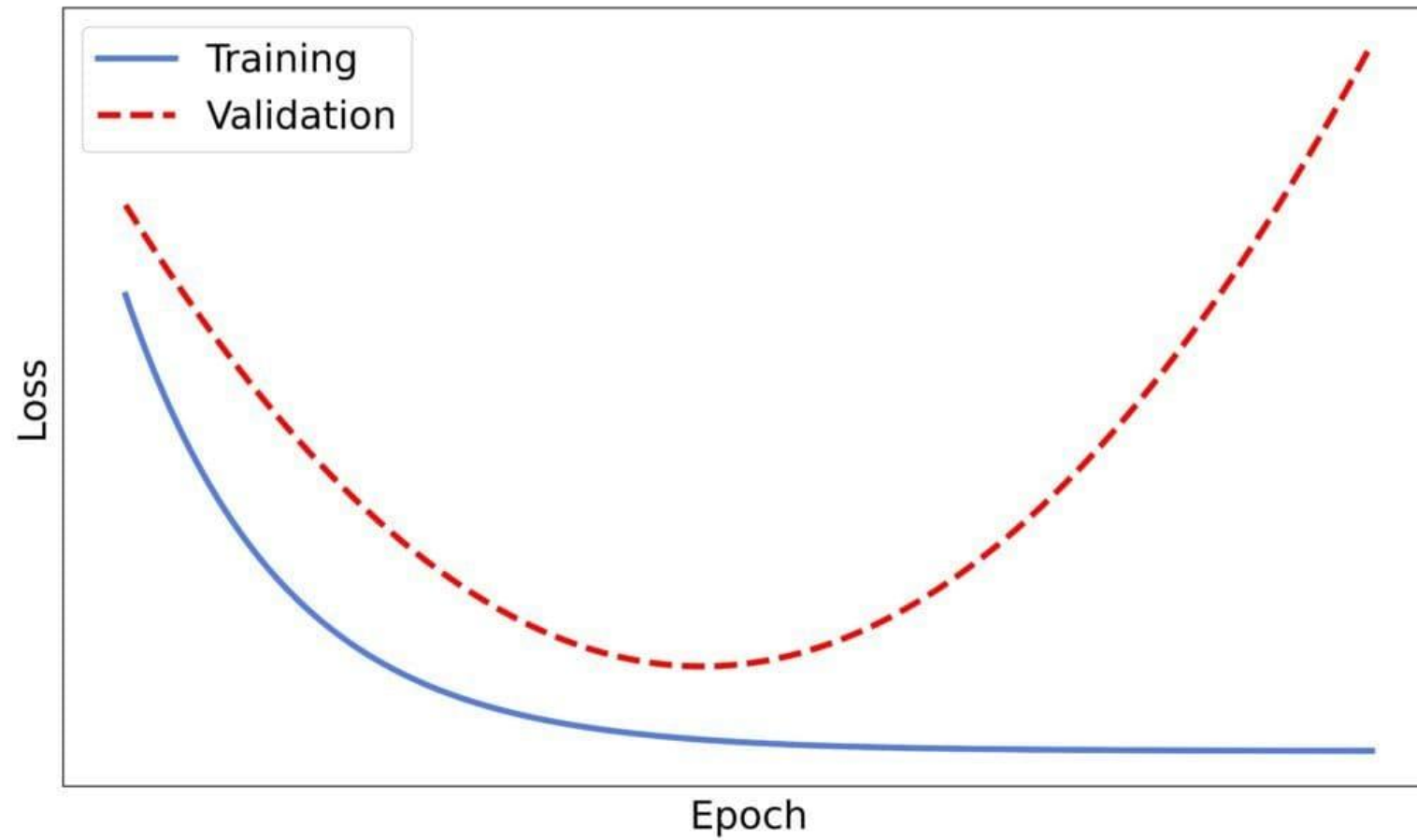
- **Sample**
- **Batch**
- **Epoch**
- **Learning rate**
- **Cost Function/Loss Function**
- **Weights/ Bias**

S1	S1
S2	S2
S3	S3
S4	S4
S5	S5
S6	S6
S7	S7
S8	S8
S9	S9
S10	S10
S11	S11
S12	S12
S13	S13
S14	S14
S15	S15
S16	S16
S17	S17
S18	S18
S19	S19
S20	S20

Batch size



Epoch

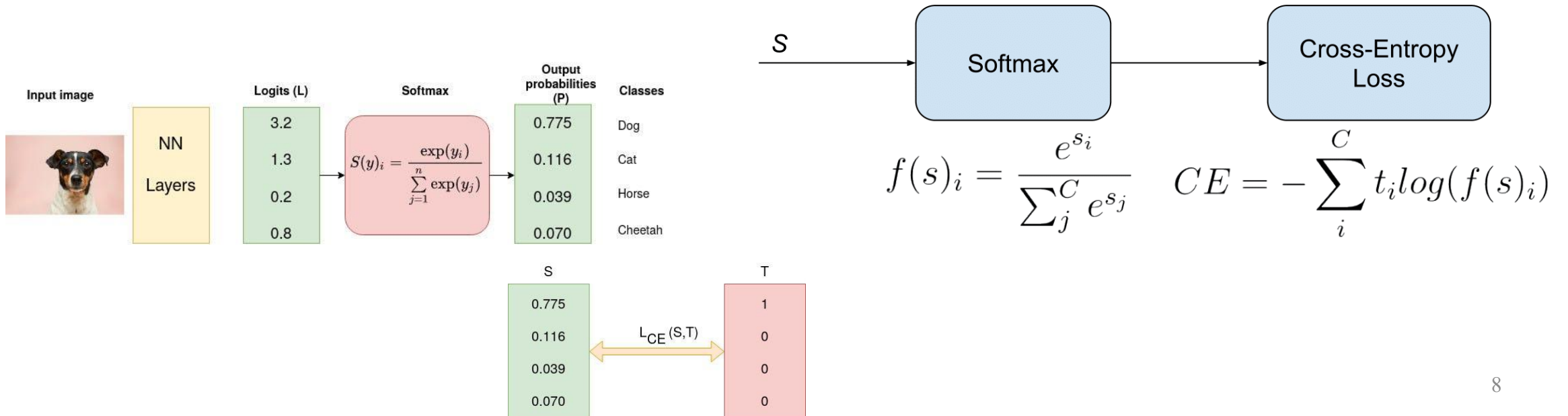


Loss function

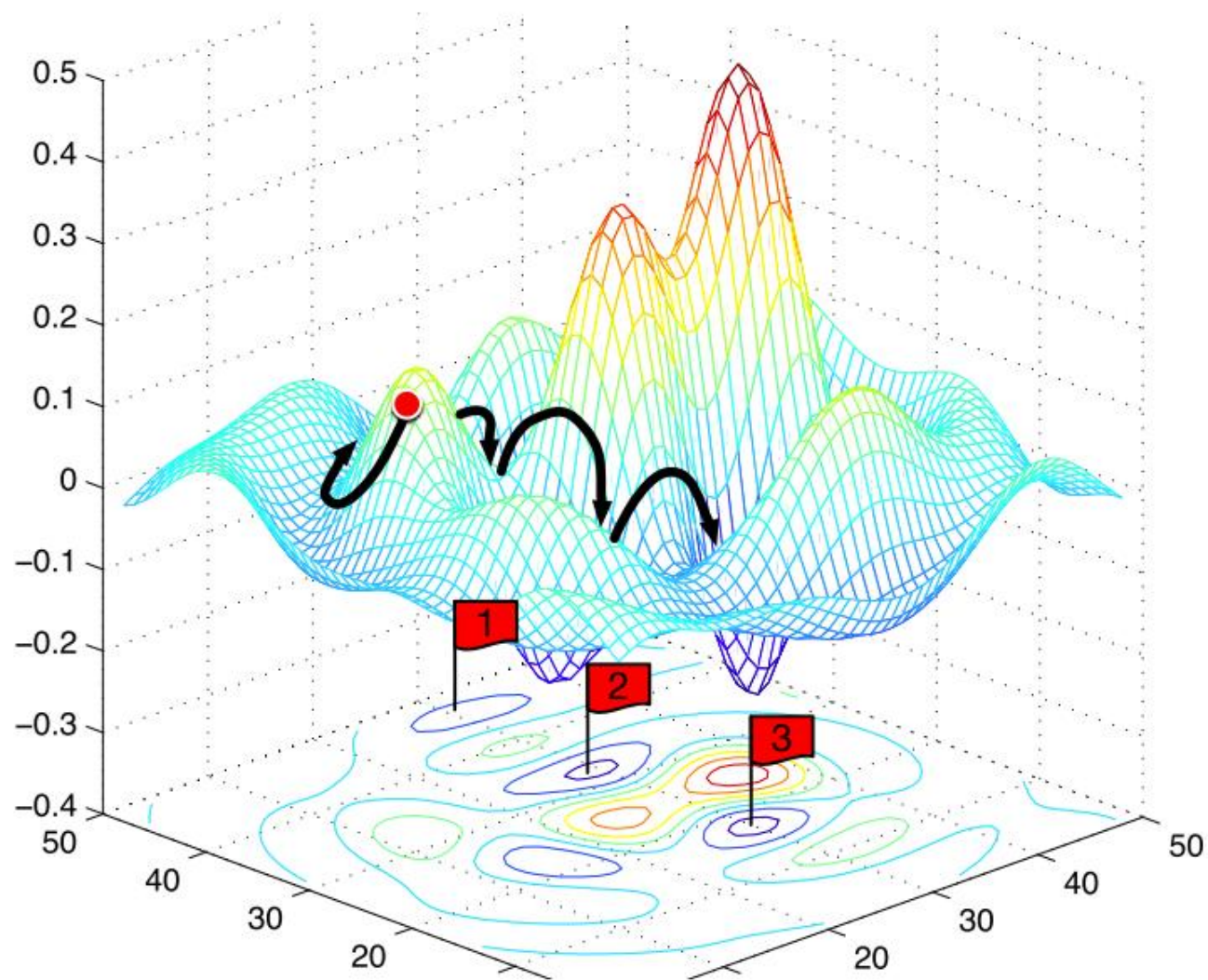
- Mean-square error loss function for regression

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- Cross entropy loss function for classification



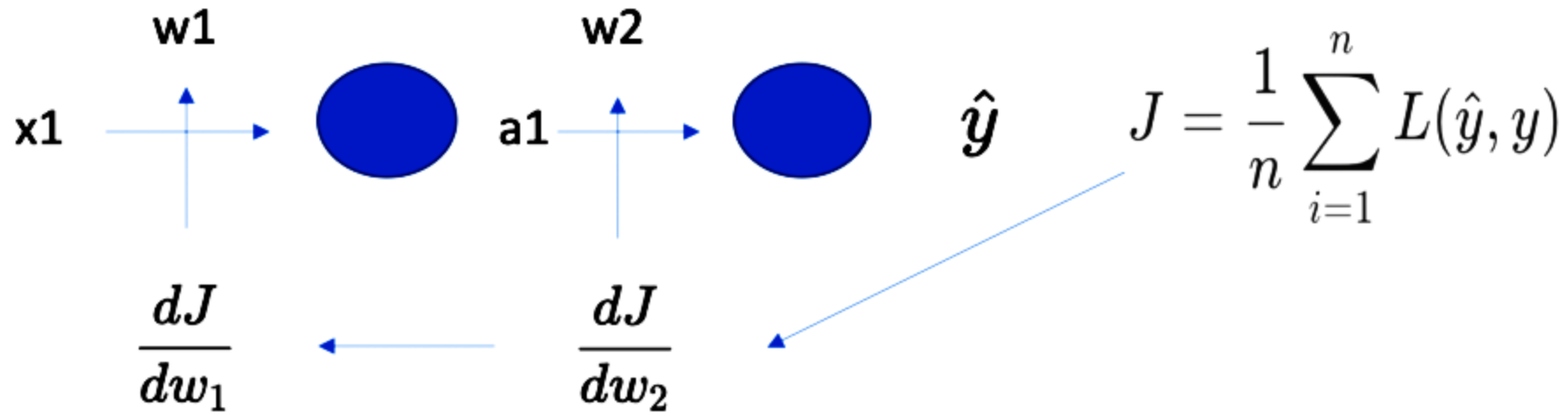
Optimizer



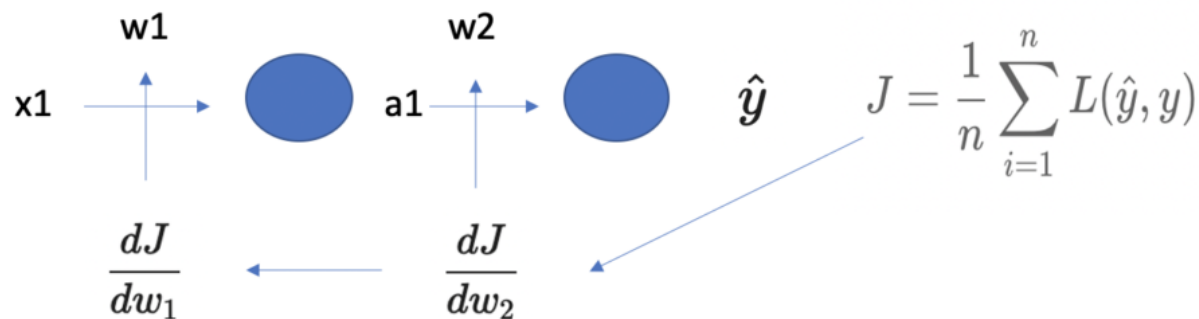
Optimizer

- SGD (Stochastic Gradient Descent)

$$z_1 = w_1 x_1 \rightarrow a_1 = \sigma(z_1) \rightarrow z_2 = w_2 a_1 \rightarrow \hat{y} = \sigma(z_2) \rightarrow J = \frac{1}{n} \sum_{i=1}^n L(\hat{y}, y)$$



SGD



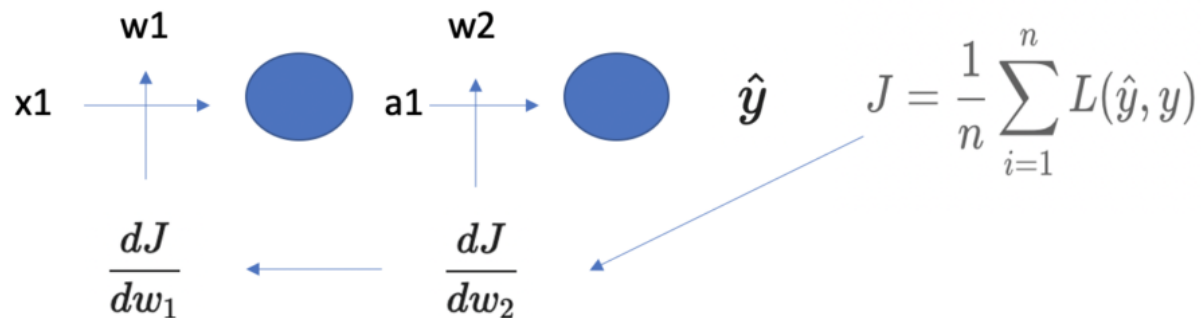
$$z_2 = w_2 a_1 \rightarrow \hat{y} = \sigma(z_2) \rightarrow J = \frac{1}{n} \sum_{i=1}^n L(\hat{y}, y)$$

$$\frac{dJ}{dw_2} = \frac{dJ}{d\hat{y}} \frac{d\hat{y}}{dz_2} \frac{dz_2}{dw_2}$$

$$\frac{d\hat{y}}{dz_2} = \frac{d}{dz_2} \sigma(z_2) = \frac{d}{dz_2} \left(\frac{1}{1 + e^{z_2}} \right)$$

$$\frac{d\hat{y}}{dz_2} = \sigma(z_2)(1 - \sigma(z_2)) = \frac{1}{1 + e^{z_2}} \left(1 - \frac{1}{1 + e^{z_2}} \right)$$

SGD



$$\frac{dz_2}{dw_2} = \frac{d}{dw_2}(w_2 a_1) = a_1$$

$$L(\hat{y}, y) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$$

$$\frac{dJ}{dw_2} = \frac{dJ}{d\hat{y}} \frac{d\hat{y}}{dz_2} \frac{dz_2}{dw_2} = \left(\frac{-y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right) [\sigma(z_2)(1 - \sigma(z_2))] a_1$$

Learning rate

$$w_{2new} = w_2 - \alpha \frac{dJ}{dw_2}$$

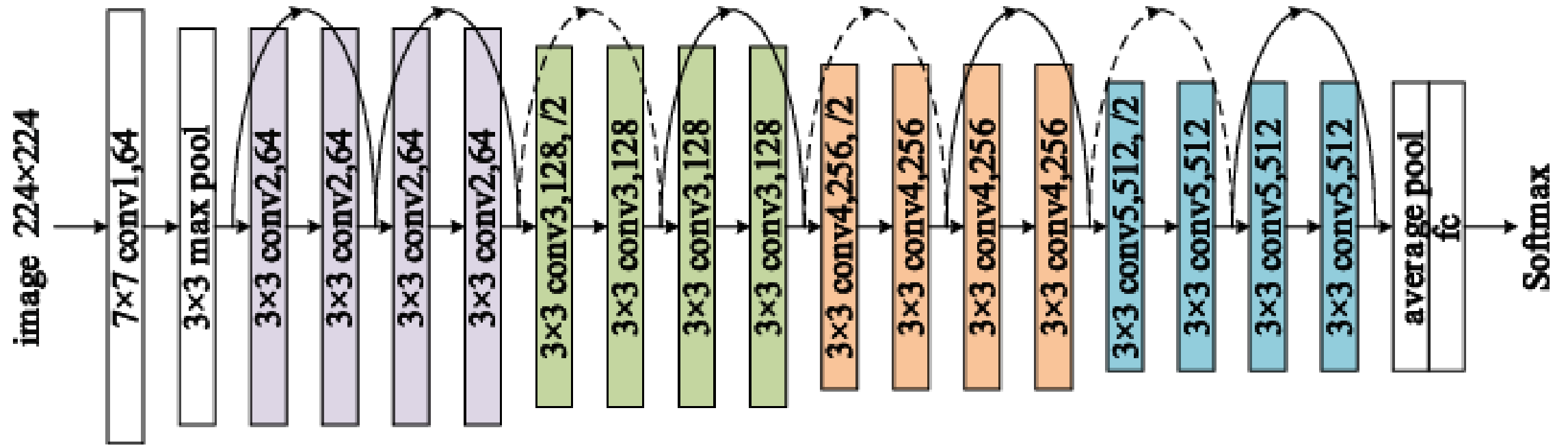
Vanishing Gradient Problem

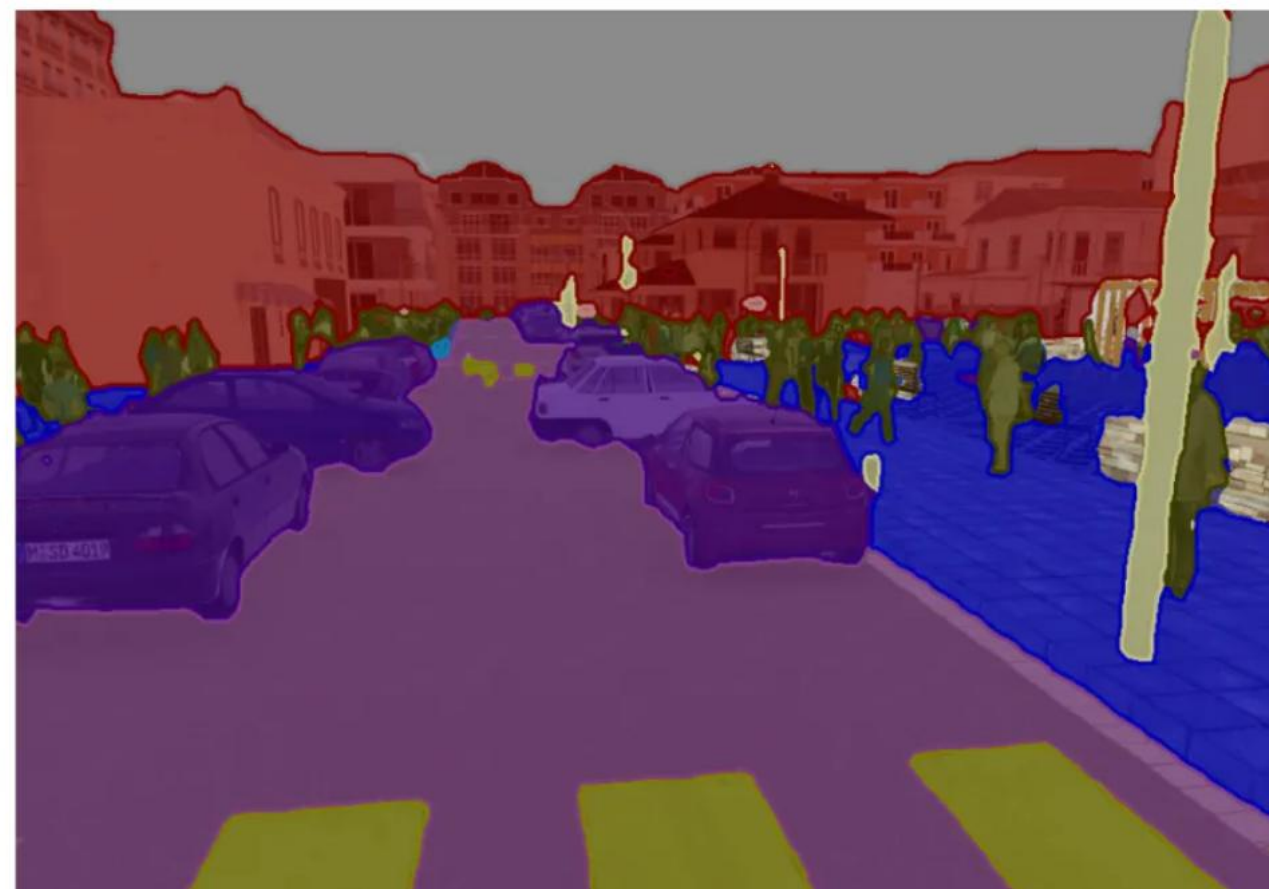
ResNet18 architecture

It takes 512 complex feature maps and boils each one down to a single average value

It drastically shrinks the number of weights the model needs to learn right at the end, which helps prevent **overfitting**

Layer Name	Output Size	ResNet-18
conv1	$112 \times 112 \times 64$	$7 \times 7, 64$, stride 2
conv2_x	$56 \times 56 \times 64$	3×3 max pool, stride 2
		$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
conv3_x	$28 \times 28 \times 128$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$
conv4_x	$14 \times 14 \times 256$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
conv5_x	$7 \times 7 \times 512$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$
average pool	$1 \times 1 \times 512$	7×7 average pool
fully connected	1000	512×1000 fully connections
softmax	1000	





Semantic Segmentation

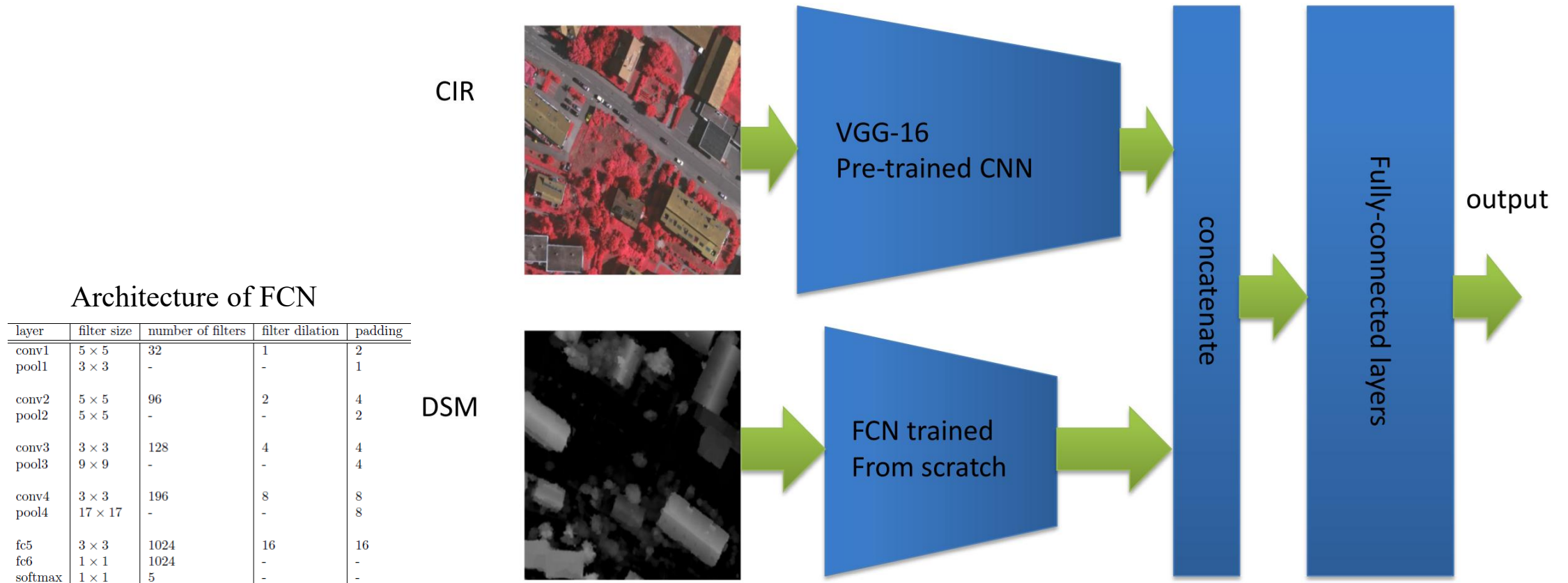
- Context information

ISPRS Semantic
Labeling Contest (2D):
Results

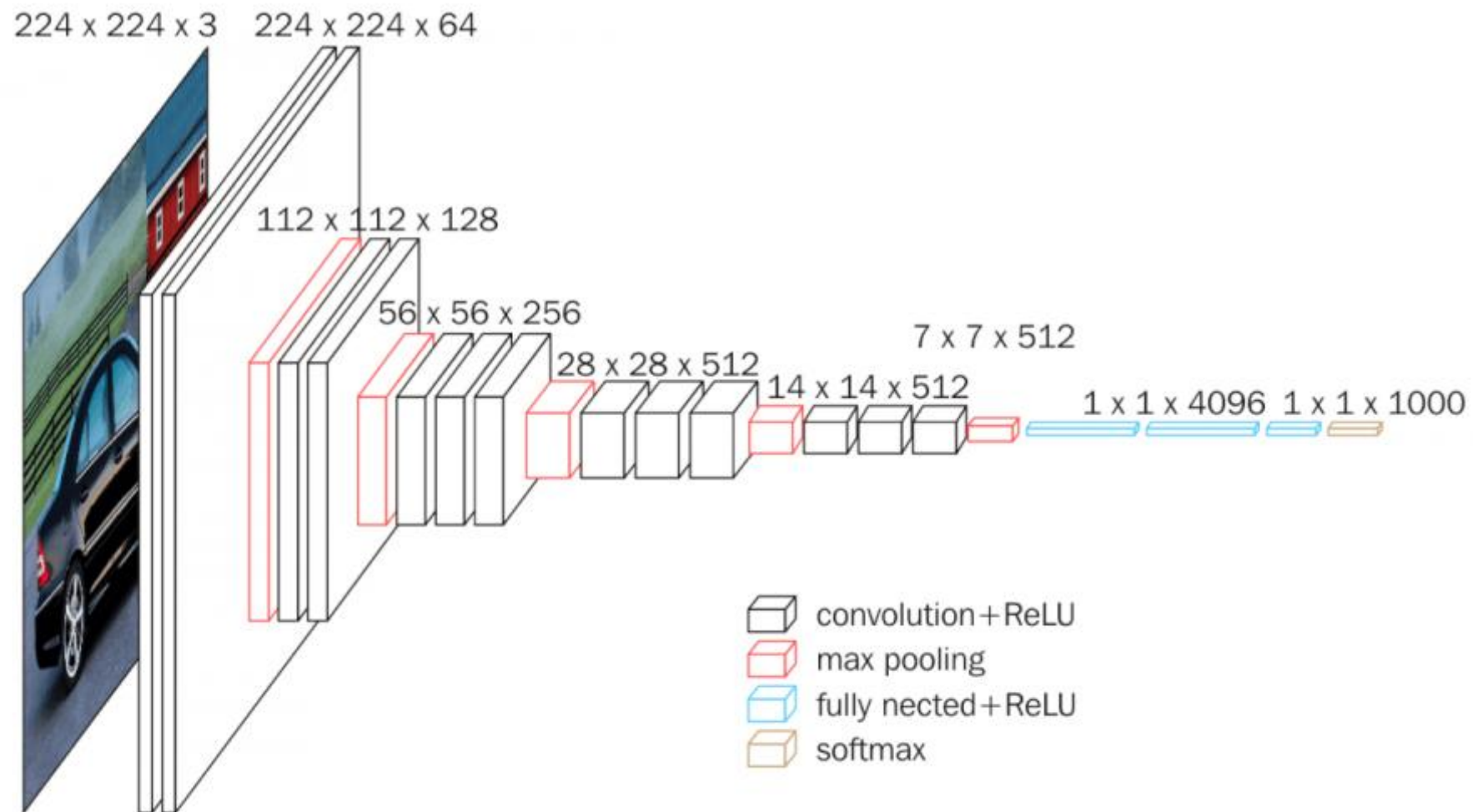
brev.	imp surf	building	low_veg	tree	car	Overall	Strategy	Details
/L_1	83.5	91.7	72.2	63.2	62.2	77.8	s	D P
/L_3	84.0	89.8	72.0	59.0	69.8	77.2	s	D P
5T_1	91.4	95.3	85.1	87.3	88.7	89.1	s	D P
5T_2	91.8	95.9	86.3	87.7	89.2	89.7	s	D P
5T_3	92.0	96.2	86.4	88.3	89.5	90.1	s	D P
5T_4	92.0	96.3	86.5	88.1	89.5	90.1	s	D P
5T_5	92.5	96.4	86.7	88.0	94.7	90.3	s	D P
5T_6	92.4	96.4	86.8	87.7	93.4	90.2	s	D P
Z_1	89.3	95.4	81.8	80.5	86.5	85.8	s	D P
3U	87.1	94.7	77.1	73.9	81.2	82.9	s	D P
RIT	91.4	96.9	84.8	86.6	92.3	88.6	s	D P
IT2	92.0	96.3	85.5	86.5	94.5	89.4	s	D P
IT3	92.3	96.8	86.5	87.3	91.3	90.0	s	D P
IT4	92.6	97.0	86.9	87.4	95.2	90.3	s	D P
IT5	90.6	95.9	83.5	83.3	93.1	87.9	s	D P
IT6	92.5	97.0	86.5	87.2	94.9	90.2	s	D P
uhZ	91.4	95.4	85.4	86.8	94.6	88.9	s	D P
\SIA	92.8	96.9	86.0	88.2	94.2	89.6	s	D P
SIA2	93.3	97.0	87.7	88.4	96.2	91.1	s	D P
SIA3	93.4	96.8	87.6	88.3	96.1	91.0	s	D P
Z1	91.4	96.1	86.1	86.6	93.3	89.2	s	D P
Z2	92.3	96.0	86.4	87.6	95.1	89.9	s	D P
Z3	93.1	96.3	87.2	88.6	96.0	90.7	s	D P
Γ_L1	88.7	91.5	82.2	82.2	90.8	85.5	s	D P
Γ_L2	89.8	94.1	83.0	82.3	90.4	86.8	s	D P
Γ_L3	89.6	93.3	83.0	82.6	91.4	86.6	s	D P
Γ_L4	89.2	93.8	82.6	81.0	91.6	86.3	s	D P
Γ_L5	90.2	93.9	83.0	82.5	91.8	86.9	s	D
Γ_L6	90.7	93.9	84.8	85.1	92.4	87.8	s	D P
Γ_L7	91.2	94.6	85.1	85.1	92.8	88.4	s	D P
ab_1	89.1	92.6	82.5	81.4	92.5	85.7	s	D P
ab_2	89.7	92.7	83.7	84.0	92.1	86.7	s	D P
ab_3	89.3	92.0	83.5	83.8	92.0	86.4	s	D P
CT_1	92.4	97.0	86.3	86.4	93.9	90.0	s	D P
S_Y1	74.6	82.1	63.8	76.9	91.2	73.8	18 s	D P
S_Y2	92.6	96.2	87.3	87.7	95.7	90.4	s	D P

Fully Convolutional Networks for Dense Semantic Labeling of High-Resolution Aerial Imagery

Abbrev.	imp surf	building	low_veg	tree	car	Overall	Strategy	Details
DST_6	92.4	96.4	86.8	87.7	93.4	90.2	s	D P



VGG-16



VGG-16

VGG16 - Structural Details													
#	Input Image			output			Layer	Stride	Kernel		in	out	Param
1	224	224	3	224	224	64	conv3-64	1	3	3	3	64	
2	224	224	64	224	224	64	conv3-64	1	3	3	64	64	
	224	224	64	112	112	64	maxpool	2	2	2	64	64	
3	112	112	64	112	112	128	conv3-128	1	3	3	64	128	
4	112	112	128	112	112	128	conv3-128	1	3	3	128	128	
	112	112	128	56	56	128	maxpool	2	2	2	128	128	
5	56	56	128	56	56	256	conv3-256	1	3	3	128	256	
6	56	56	256	56	56	256	conv3-256	1	3	3	256	256	
7	56	56	256	56	56	256	conv3-256	1	3	3	256	256	
	56	56	256	28	28	256	maxpool	2	2	2	256	256	
8	28	28	256	28	28	512	conv3-512	1	3	3	256	512	
9	28	28	512	28	28	512	conv3-512	1	3	3	512	512	
10	28	28	512	28	28	512	conv3-512	1	3	3	512	512	
	28	28	512	14	14	512	maxpool	2	2	2	512	512	
11	14	14	512	14	14	512	conv3-512	1	3	3	512	512	
12	14	14	512	14	14	512	conv3-512	1	3	3	512	512	
13	14	14	512	14	14	512	conv3-512	1	3	3	512	512	
	14	14	512	7	7	512	maxpool	2	2	2	512	512	
14	1	1	25088	1	1	4096	fc		1	1	25088	4096	
15	1	1	4096	1	1	4096	fc		1	1	4096	4096	
16	1	1	4096	1	1	1000	fc		1	1	4096	1000	
Total													138,423,208

$(3 \times 3 \times 3 + 1) \times 64$
 $(3 \times 3 \times 64 + 1) \times 64$
 0

$(25088 + 1) \times 4096$
 $(4096 + 1) \times 4096$
 $(4096 + 1) \times 1000$

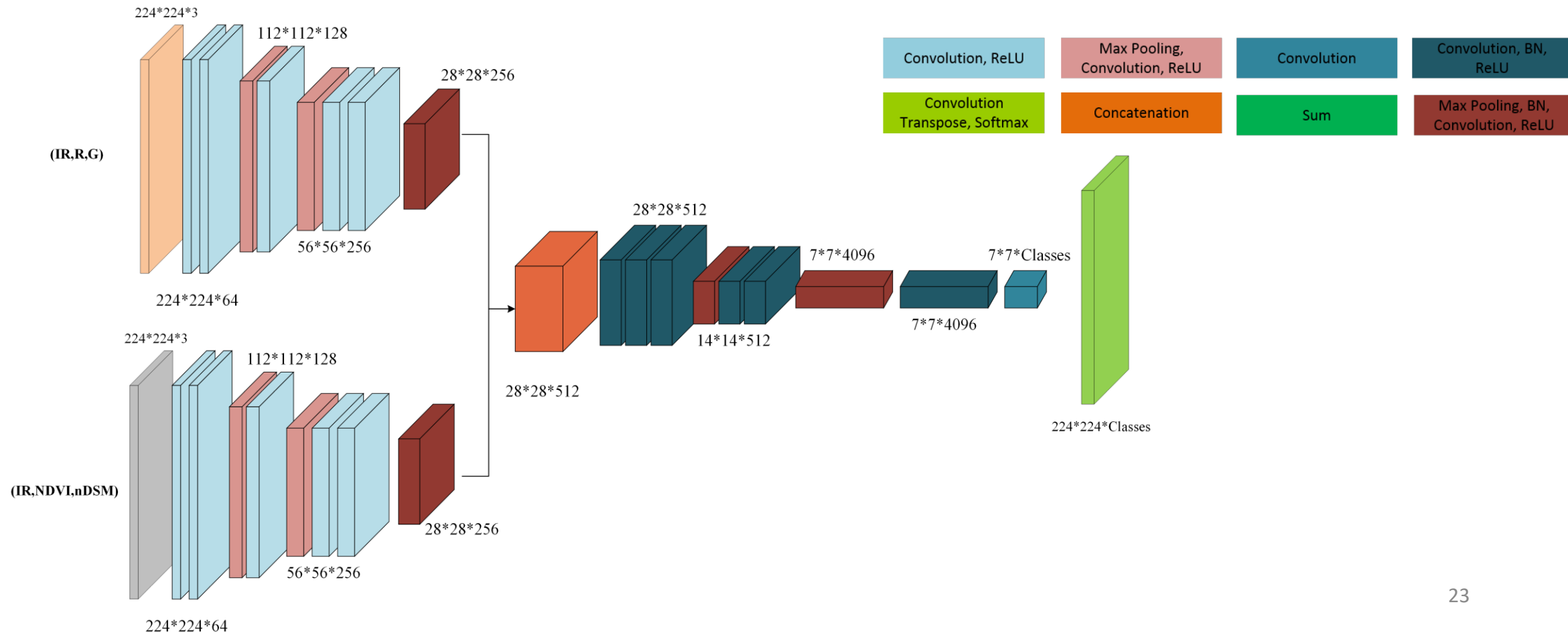
VGG-16

VGG16 - Structural Details													
#	Input Image			output			Layer	Stride	Kernel		in	out	Param
1	224	224	3	224	224	64	conv3-64	1	3	3	3	64	1792
2	224	224	64	224	224	64	conv3-64	1	3	3	64	64	36928
	224	224	64	112	112	64	maxpool	2	2	2	64	64	0
3	112	112	64	112	112	128	conv3-128	1	3	3	64	128	73856
4	112	112	128	112	112	128	conv3-128	1	3	3	128	128	147584
	112	112	128	56	56	128	maxpool	2	2	2	128	128	65664
5	56	56	128	56	56	256	conv3-256	1	3	3	128	256	295168
6	56	56	256	56	56	256	conv3-256	1	3	3	256	256	590080
7	56	56	256	56	56	256	conv3-256	1	3	3	256	256	590080
	56	56	256	28	28	256	maxpool	2	2	2	256	256	0
8	28	28	256	28	28	512	conv3-512	1	3	3	256	512	1180160
9	28	28	512	28	28	512	conv3-512	1	3	3	512	512	2359808
10	28	28	512	28	28	512	conv3-512	1	3	3	512	512	2359808
	28	28	512	14	14	512	maxpool	2	2	2	512	512	0
11	14	14	512	14	14	512	conv3-512	1	3	3	512	512	2359808
12	14	14	512	14	14	512	conv3-512	1	3	3	512	512	2359808
13	14	14	512	14	14	512	conv3-512	1	3	3	512	512	2359808
	14	14	512	7	7	512	maxpool	2	2	2	512	512	0
14	1	1	25088	1	1	4096	fc		1	1	25088	4096	102764544
15	1	1	4096	1	1	4096	fc		1	1	4096	4096	16781312
16	1	1	4096	1	1	1000	fc		1	1	4096	1000	4097000
Total													138,423,208

Supervised Classification of Multisensor Remotely Sensed Images Using a Deep Learning Framework

Early fusion

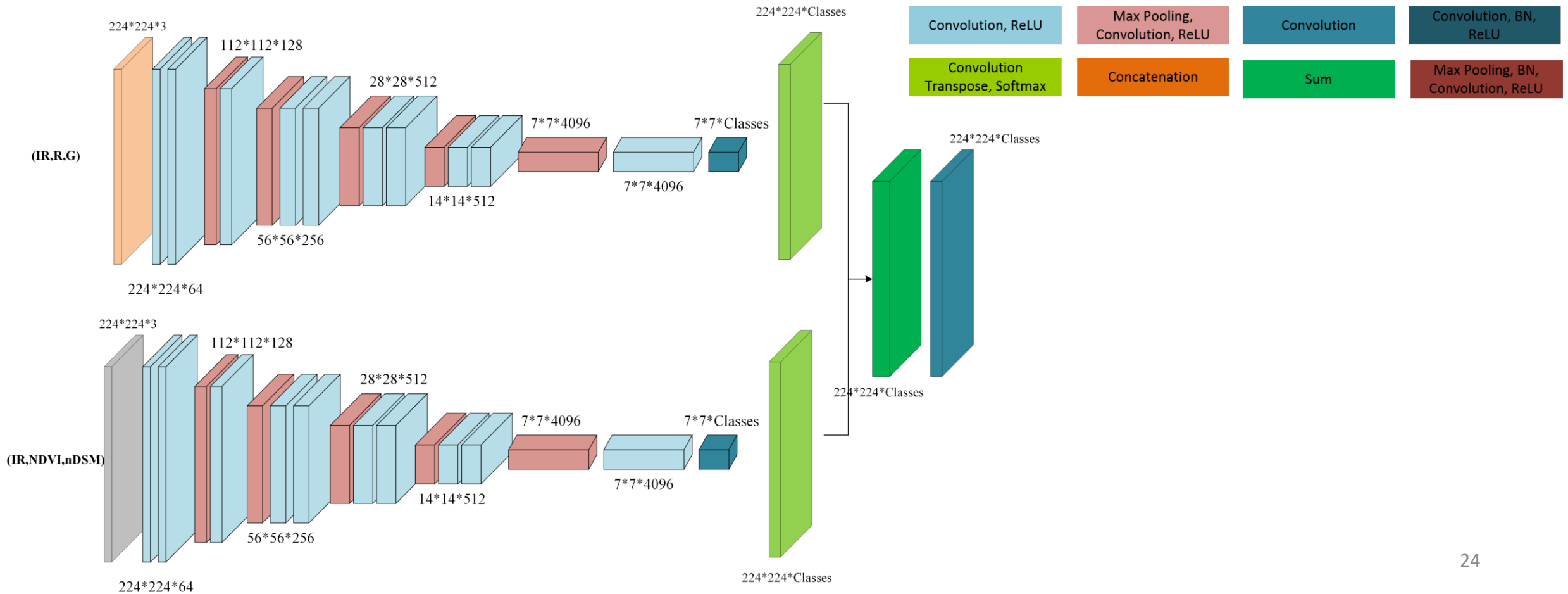
Abbrev.	imp surf	building	low_veg	tree	car	Overall	Strategy	Details
RIT6	92.5	97.0	86.5	87.2	94.9	90.2	s	D P



Supervised Classification of Multisensor Remotely Sensed Images Using a Deep Learning Framework

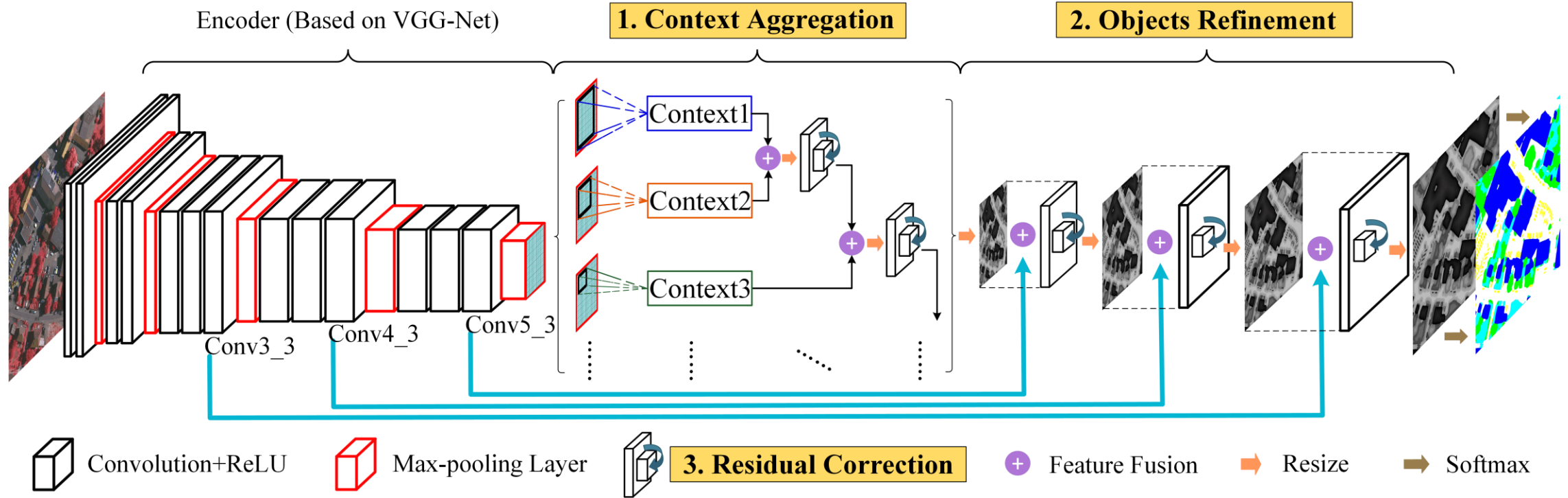
Late fusion

Abbrev.	imp surf	building	low_veg	tree	car	Overall	Strategy	Details
RIT6	92.5	97.0	86.5	87.2	94.9	90.2	s	D P



Semantic Labeling in Very High Resolution Images via A Self-Cascaded Convolutional Neural Network

Abbrev.	imp surf	building	low_veg	tree	car	Overall	Strategy	Details
CASIA3	93.4	96.8	87.6	88.3	96.1	91.0	s	D P



Question and
feedback on
the assignment

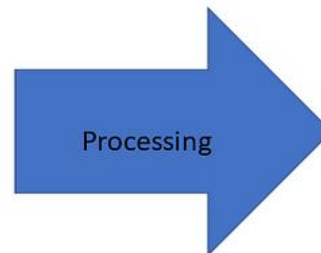


Dataset preparation

- Using TensorFlow, generate 5 **TFRecord** files
- Shuffle Data

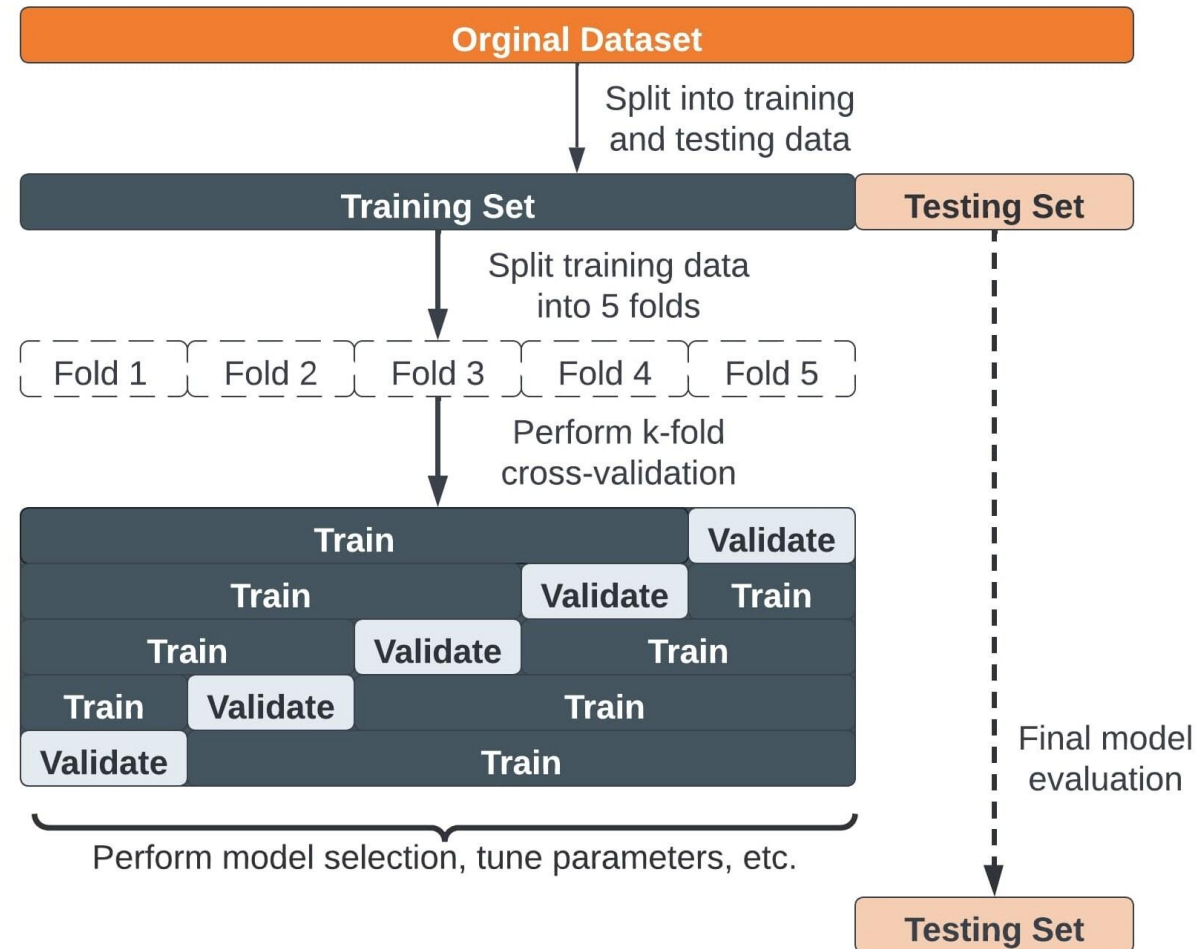
TFRecord

- A serialized binary format
- Faster Data Loading
- Easy to preprocess
- Supports a wide range of data types
- Handling Large Datasets
- Distributed Training
- Custom Data Formats



TFRecord

K-Fold cross validation





Any
Questions